INDIAN INSTITUTE OF TECHNOLOGY

EE-3202

Power System II

Experiment-2

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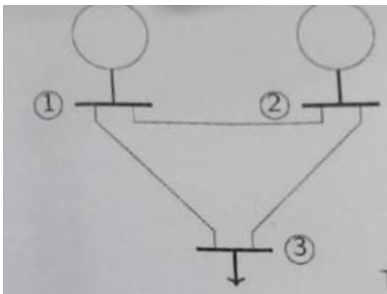
Gauss Siedel Method

$$Y_{11} = \boxed{y_{12} + y_{13} + y'_{12} + y'_{13}}$$

$y_{12} \rightarrow \frac{1}{X_{12}}$, $y_{13} \rightarrow \frac{1}{X_{13}}$, $y'_{12} \rightarrow \frac{Y'_{12}}{2}$, $y'_{13} \rightarrow$ ✓

$$Y_{12} = \boxed{(-y_{12})} = Y_{21}$$

Aim-



Bus Code	R (p.u.)	X (p.u.)	$\frac{Y}{2}$ (p.u.)
1-2	0	j0.1	j0.01
1-3	0	j0.1	j0.01
2-3	0	j0.1	j0.01

Table: Bus data

Bus	P_g	Q_g	P_d	Q_d	$\|V\|$	δ	Type
1	—	—	—	—	1	0	Slack
2	0.6661	—	—	—	1.05	—	PV
3	—	—	2.8653	1.2244	—	—	PQ

Assume a flat voltage start, determine the voltage at the end of first iteration using G-S method. $0.2 \leq Q_2 \leq 2$. Take $\alpha = 1.6$.

reciprocal is off diagonal waale

Manual Solution-

$y_{12} = \frac{1}{j0.1} = (-j10)$ | $y_{13} = \frac{1}{j0.1} = (-j10)$ | $Y_{11} = \left(y_{12} + \frac{y_{12}'}{2} + y_{13} + \frac{y_{13}'}{2}\right)$

$\frac{y_{12}'}{2} = j0.01$ | $\frac{y_{13}'}{2} = -j0.01$ | $Y_{11} = -j(19.98)$

$Y_{12} = -y_{12} = j10$

$$\therefore \; Y_B = \begin{bmatrix} -j19.98 & j10 & j10 \\ j10 & -j19.98 & j10 \\ j10 & j10 & -j19.98 \end{bmatrix}$$

Iteration-1:

(i) Bus-2 : (PV bus)

→ $|V_2|_{spec} = 1.05$, $Q_2 \in [0.2, 2]$

→ $V_1 = (1 + j0)$, $V_2(0) = \boxed{1.05 + j0}$, $V_3(0) = 1 + j0$, $(P_2)_{sch} = 0.6661$

→ $Q_2(1) = -Im[1.05 \times (-j0.979)] = (+1.02795)$ pu

→ $\because$ $1.02 \in [0.2, 2]$ ⇒ Proceed.

→ $V_{2,temp}(1) = (0.634 - j20.97) = (1.05 + j0.03)$ pu

$V_{2,temp,acc}(1) = \boxed{1.05} + 1.6[(1.05 + j0.03) - \boxed{1.05}]$

$= (1.05 + j0.05)$ pu

$V_{0,temp} - V_1(1) = (1.048 + j0.05) = (1.05 \angle 2.77)$ pu

(ii) Bus-3 : (PQ-bus)

$V_2(1) = (1.048 + j0.05)$

$P_{3,sch} - jQ_{3,sch} = (-2.86 + j1.22)$

$V_{3,temp}(1) = (-2.35 - j19.26) = (0.96 - j0.11)$ pu

$V_{final} = (0.94 - j0.18) = (0.96 \angle -11.32°)$ pu

→ $\therefore$ $V_1(1) = (1 \angle 0°)$ pu

$V_2(1) = (1.05 \angle 2.77°)$ pu

$V_3(1) = (0.96 \angle -11.32°)$ pu

Code Solution-

```
alpha = 1.6;
j = 1i;
X = j*0.1;
B = j*0.01; %(Y/2)
y = 1/X;

Y = zeros(3);

Y(1,1) = y+y + B+B;
Y(2,2) = y+y + B+B;
Y(3,3) = y+y + B+B;
Y(1,2) = -y; Y(2,1) = -y;
Y(1,3) = -y; Y(3,1) = -y;
Y(2,3) = -y; Y(3,2) = -y;

Pg = [0;0.6661;0];
Qg = [0;0;0];
Pd = [0;0;2.8653];
Qd = [0;0;1.2244];
P = Pg - Pd;
Q = Qg - Qd;

V = [1+0i;1.05+0i;1+0i];    %start
Qmin = 0.2; Qmax = 2;

Q2 = -imag(conj(V(2))*(Y(2,:)*V));
if Q2 < Qmin
    Q2 = Qmin;
elseif Q2 > Qmax
    Q2 = Qmax;
end

V2_old = V(2);
V(2) = (1/Y(2,2))*((P(2)-1i*Q2)/conj(V(2))- Y(2,1)*V(1) - Y(2,3)*V(3));
V(2) = V2_old + alpha*(V(2)-V2_old);
V(2) = 1.05*exp(1i*angle(V(2)));

V3_old = V(3);
V(3) = (1/Y(3,3))*((P(3)-1i*Q(3))/conj(V(3)) - Y(3,1)*V(1) - Y(3,2)*V(2));
V(3) = V3_old + alpha*(V(3)-V3_old);

disp('Voltage after first iteration')
for i=1:3
    fprintf('Bus %d : |V| = %.4f  angle = %.4f deg\n', (i), (abs(V(i))), (angle(V(i))*180/pi));
end
```

Output-

```
Voltage after first iteration
Bus 1 : |V| = 1.0000  angle = 0.0000 deg
Bus 2 : |V| = 1.0500  angle = 2.7699 deg
Bus 3 : |V| = 0.9613  angle = -11.3273 deg
```